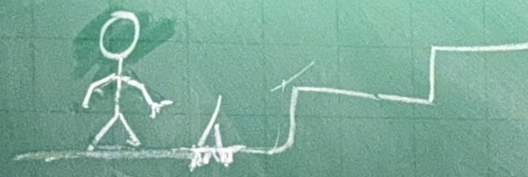
16 PDF
22L DWF



→ MKV

BDS
FBX
PNG
TIF
OKF
DWF

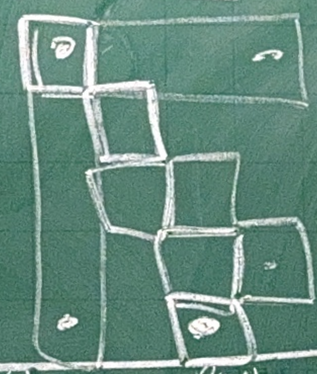
→ Slides

→ prática - explicação

→ Intervalo

Ajuda de ?

1	
2	
3	Hoje 25/03
4	01/04
5	08/04



TG { S, R

Ponto

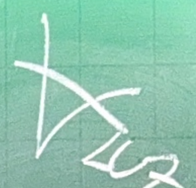
$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}_{1\times4} \times \begin{bmatrix} \times \end{bmatrix}_{4\times4} = \begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix}$$

$$\begin{bmatrix} 3 \end{bmatrix}_{4\times4} \times \begin{bmatrix} \cdot \end{bmatrix}_{4\times4} = \begin{bmatrix} \cdot \end{bmatrix}_{4\times4}$$

Comp. Gráfica → Pipeline Gráfico
OpenGL

→ { Dados
Rotinas

Espaço gráfico



DADOS

→ Dimensão → 2D
1D, 2D, 3D, 4D ... ND
$2D\frac{1}{2}$

→ Tipo { Raster
Vetorial

→ Sistema de Referência Cartesiano

OpenGL → Animations Gráfica

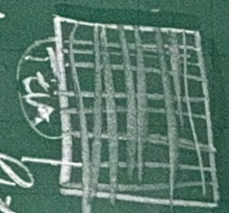
Pixel resolução cores

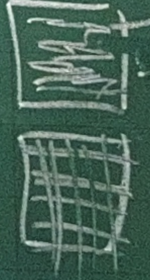
Saída Raster

Cinema

Buffer

GPU

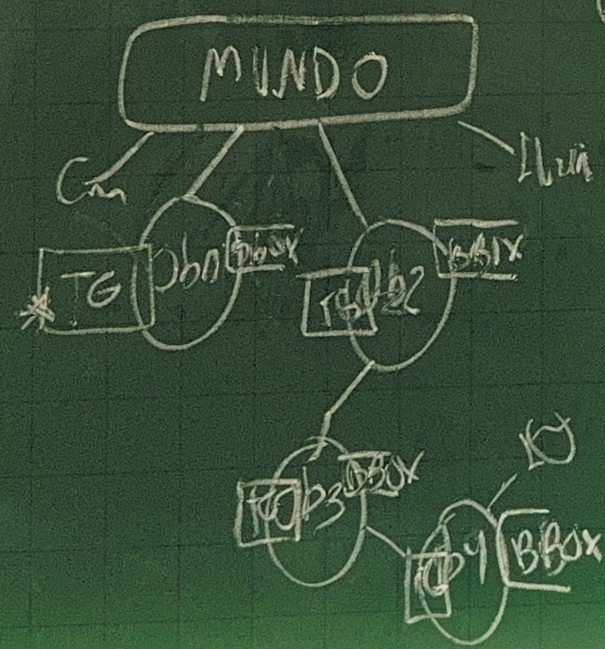


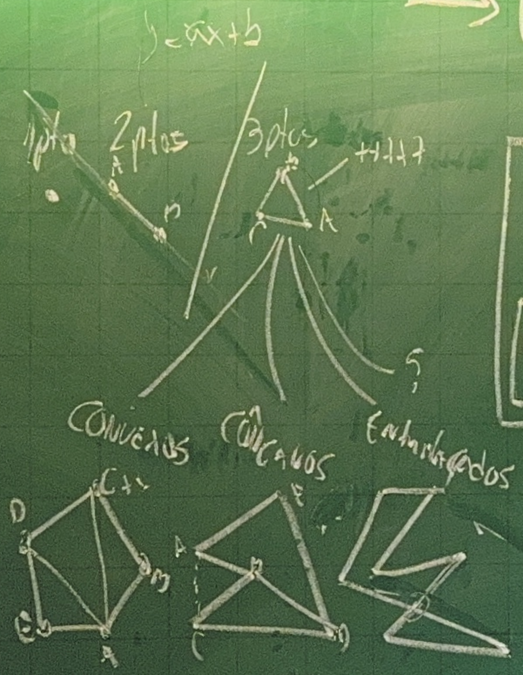


Cena Gráfica

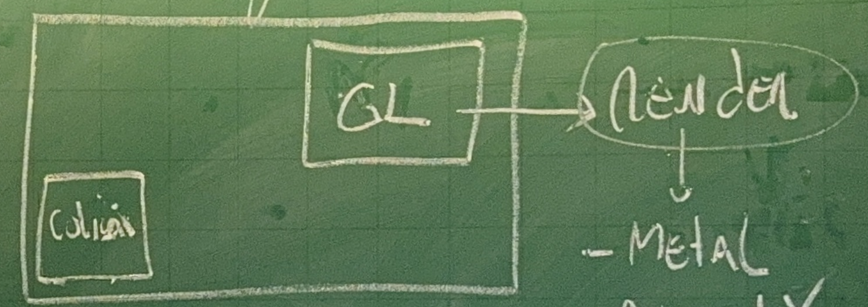
Popular Obj. Gráficos
- Forma
- Aparência
- Movto

Formalizam
Grafo de Cena



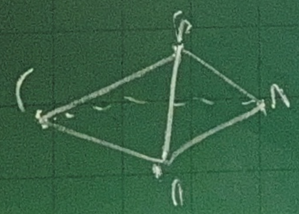


→ Motores de Jogos
Engine



- Metal
- DirectX
- Vulkan
- OpenGL → OpenTK (C#)

GLUT

GLU — Padrão Câmaras

GL → Base